Ch 7.4: Cubic splines

Lecture 22 - CMSE 381

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Announcements

Last time:

- 7.2 Step functions
- 7.3 Basis functions

This lecture:

- 7.4 Cubic splines

Announcements:

- Homework # 7 is now (for real) on github

Section 1

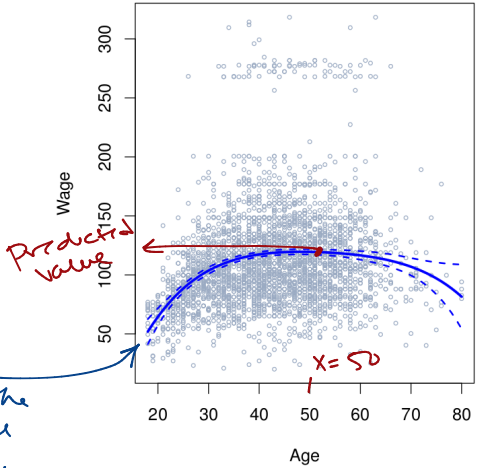
Last time

Polynomial regression

X= age p= 1

Y= wage

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$



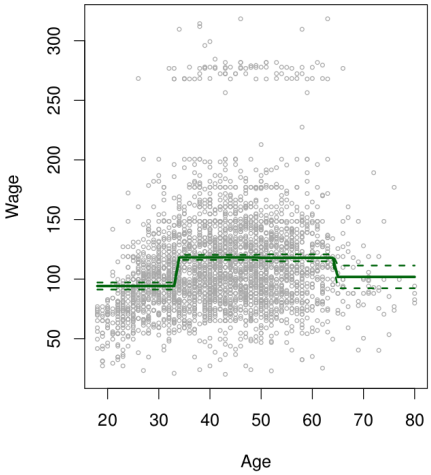
draw the model function

Step function regression

$c_1 \quad c_2 \quad c_3 \quad \cdots \quad c_K$

$$
\begin{aligned}
C_0(X) &= I(X < c_1), \\
C_1(X) &= I(c_1 \le X < c_2), \\
C_2(X) &= I(c_2 \le X < c_3), \\
&\vdots \\
C_{K-1}(X) &= I(c_{K-1} \le X < c_K), \\
C_K(X) &= I(c_K \le X),
\end{aligned}
$$

$$y_i = \beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i) + \varepsilon_i$$

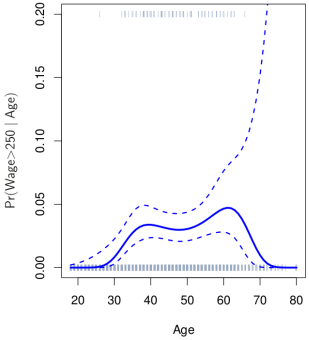


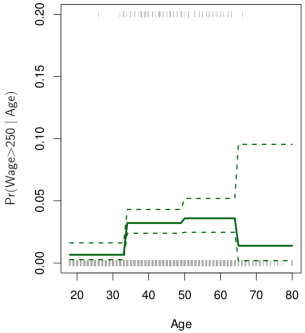
Classification version

$$\Pr(y_i > 250 \mid x_i) =$$

$$\frac{\exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}{1 + \exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}$$

$$\frac{\exp(\beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i))}{1 + \exp(\beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i))}$$





Basis Functions Setup

Polynomial and piecewise-constant regression models are special cases of a *basis function* approach.

$p = 1$

$$y_i = \beta_0 + \beta_1 \underbrace{b_1(x_i)} + \beta_2 \underbrace{b_2(x_i)} + \cdots + \beta_K \underbrace{b_K(x_i)} + \varepsilon_i$$

Poly: $b_i(x) = x^i$

Step fcn: $b_i(x) = C_i(x)$

Goal:
- Cubic splines
- This is a version of this basis function idea.

Section 2

Regression Splines

Piecewise polynomials

- Fit a polynomial regression

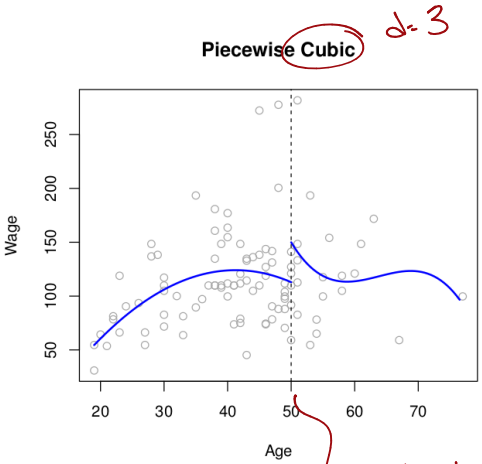
$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$

- Let the β_i's be different at different locations of the range.

$$f(x) = \begin{cases} \beta_0^1 + \beta_1^1 x + \beta_2^1 x^2 & x \in \text{part of range} \\ \beta_0^2 + \beta_1^2 x + \beta^2 x^2 & x \in \text{some other range} \\ \vdots & \end{cases}$$

- Points where the coefficients change are called "knots"
- If no knots, a piecewise polynomial is simply a polynomial.
- Step functions are piecewise polynomials of degree $d=0$

Example of piecewise polynomial



d = 3

Problem: This is a discontinuous function.

Example:

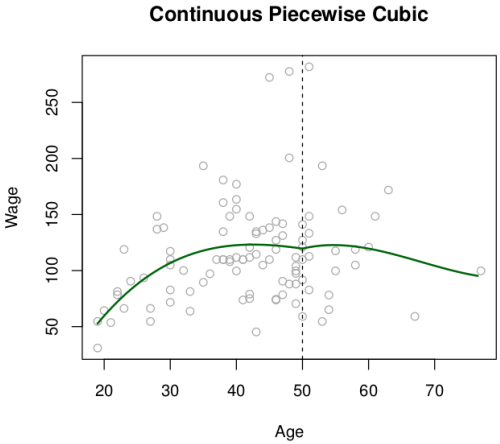
$$y_i = \begin{cases} \beta_{01} + \beta_{11}x_i + \beta_{21}x_i^2 + \beta_{31}x_i^3 + \epsilon_i & \text{if } x_i < c \\ \beta_{02} + \beta_{12}x_i + \beta_{22}x_i^2 + \beta_{32}x_i^3 + \epsilon_i & \text{if } x_i \geq c. \end{cases}$$

8 degrees of freedom.

One knot at c = 50

The fix

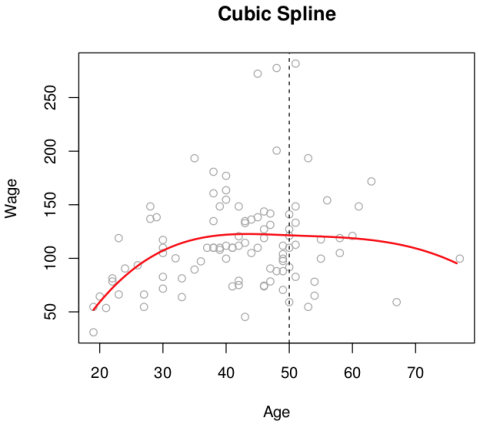
- Fit piecewise polynomial
- Require continuity at knots



The better fix: Cubic splines

d = 3

- Fit piecewise Cubic polynomial
- Require continuity at knots
- Require the first and second derivatives to be continuous at knots

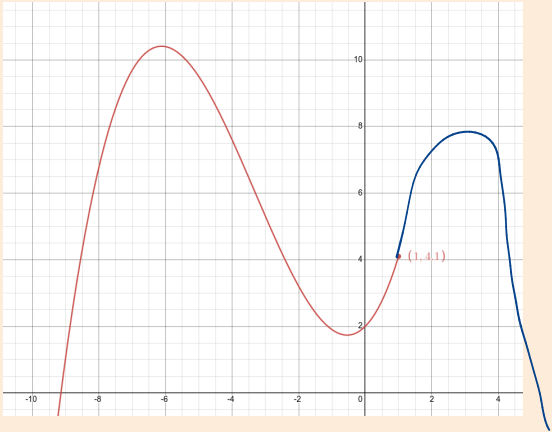


Example

We have the following piecewise cubic polynomial:

$$f(x) = \begin{cases} 2 + x + x^2 + 0.1x^3 & x \leq 1 \\ b_0 + b_1 x + b_2 x^2 - x^3 & x > 1 \end{cases}$$

What are b_1, b_1, and b_2 to make this a cubic spline?



Check your answers: desmos.com/calculator/ns4tr7mw0n

More space for work

$$f(x) = \begin{cases} \overset{f_1}{2} + x + x^2 + 0.1x^3 & x \le 1 \\ \underset{f_2}{3.1} + -2.3x + 4.3x^2 - x^3 & x > 1 \end{cases}$$

b_0 b_1 b_2

$b_0 = 3.1$

$f_2(1) = f_1(1)$

$b_0 + b_1 + b_2 - 1 = 2 + 1 + 1 + 0.1$

$f_2'(1) = f_1'(1)$

$b_1 + 2b_2 - 3 = 1 + 2 + 0.3$

$b_1 = -2.3$

$f_2''(1) = f_1''(1)$

$b_2 = 4.3 \Leftarrow 2b_2 - 6 = 2 + 0.6$

$f_2(x) = b_0 + b_1 x + b_2 x^2 - x^3$

$f_2'(x) = b_1 + 2b_2 x - 3x^2 \Leftarrow$

$f''(x) = 2b_2 - 6x$

$f_1(x) = 2 + x + x^2 + 0.1x^3$

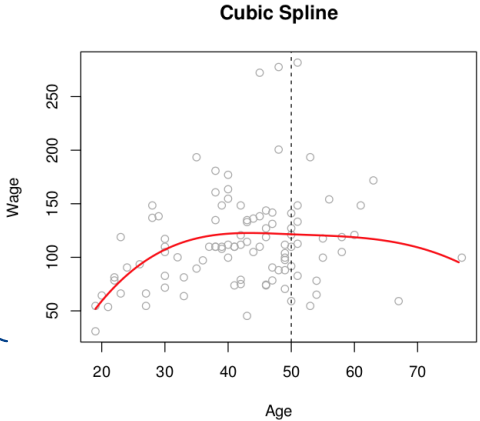
$f_1'(x) = 1 + 2x + 0.3x^2 \Leftarrow$

$f_1''(x) = 2 + 0.6x$

Cubic splines: degrees of freedom

1 knot
$d=3$

$$f(x) = \begin{cases} \beta_0^1 + \beta_1^1 x + \beta_2^1 x^2 + \beta_3^1 x^3 & x < c \\ \beta_0^2 + \beta_1^2 x + \beta_2^2 x^2 + \beta_3^2 x^3 & x > c \end{cases}$$



- 8 original degrees of freedom
- Three constraints

$\Rightarrow 8 - 3 = 5$ degrees of freedom

K knots $\rightsquigarrow$ $4 + K$ degrees of freedom $d=3$

Spline basis representation

Want to pick b_i so that we represent a cubic spline with K knots as

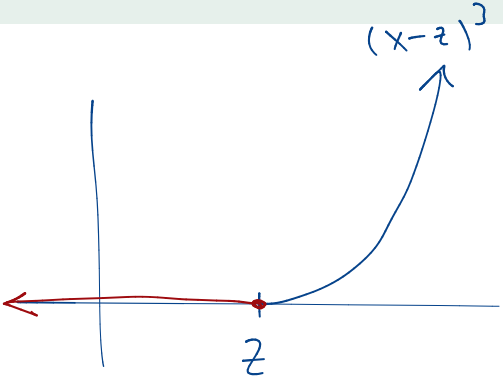
$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_{K+3} b_{K+3}(x_i) + \varepsilon_i$$

Truncated power basis function

z = knot

$$h(x,z) = (x-z)_+^3 = \begin{cases} (x-z)^3 & \text{if } x > z \\ 0 & \textit{else} \end{cases}$$

Desmos link: https://www.desmos.com/calculator/esucuulbgj



The basis for cubic splines

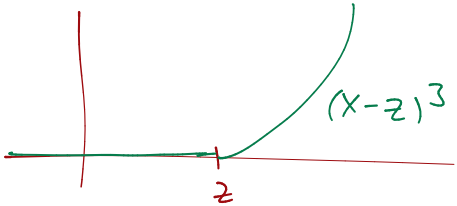
Given knots at $\underbrace{z_1, \cdots, z_K}$

- X
- X^2
- X^3
- $h(X, z_1)$
- $h(X, z_2)$
- $\vdots$
- $h(X, z_K)$

3

K

degrees of freedom = K+4

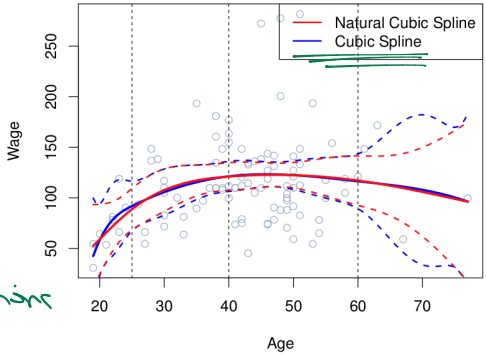


$$f(X) = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 h(X, z_1) + \beta_5 h(X, z_2) + \cdots + \beta_{k+3} h(X, z_K)$$

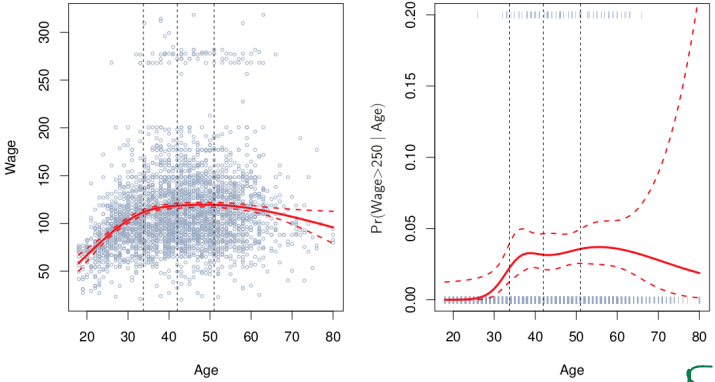
Coding example

Notes on cubic splines

- To the human eye, we can't detect the 3rd derivative discontinuities
  Reasonable option for fns
- Other choices of basis functions to still get cubic splines



Where to put the knots?



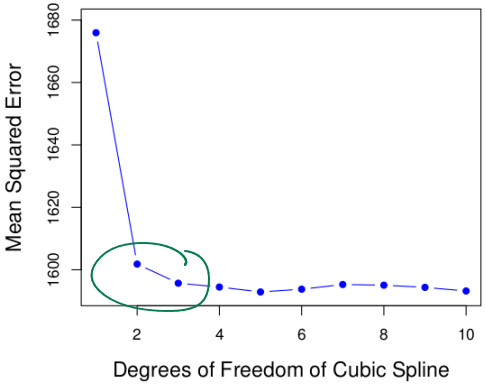
Plan A:
Evenly spaced

Plan B:
Data driven version,
Put at quartiles
In this example

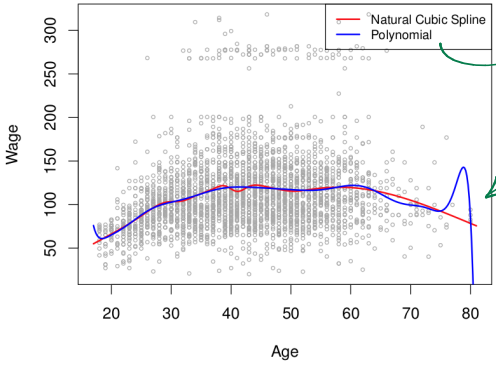
How many knots to use?

When in doubt, Cross-Validate.

Pics: 10-fold CV
for each choice of
# of knots



Cubic splines vs Polynomial Regression



one global polynomial

tend to have very wiggly boundaries

Cubic splines tend to get rid of these issues

Next time

20	F	Nov 4	Polynomial & Step Functions.	7.1,7.2	
21	M	Nov 7	Step Functions	7.2	
22	W	Nov 9	Basis functions, Regression Splines	7.3,7.4	
23	F	Nov 11	Decision Trees	8.1	HW #7 Due
24	M	Nov 14	Ensemble methods	8.2	
25	W	Nov 16	Maximal Margin Classifier	9.1	
26	F	Nov 18	SVC	9.2	HW #8 Due
27	M	Nov 21	SVM	9.3, 9.4, 9.5	
28	W	Nov 23	Single layer NN	10.1	
	F	Nov 25	No class - Thanksgiving		
29	M	Nov 28	Multi Layer NN	10.2	HW #9 Due
30	W	Nov 30	CNN	10.3	
31	F	Dec 2	Unsupervised Learning & Clustering	12.1, 12.4	
32	M	Dec 5	More Clustering	12.4	HW #10 Due
	W	Dec 7	Review		
	F	Dec 9	Midterm #3	Bring your cheat sheet and a non-internet-connected calculator	